



Complex Social &
Computational Systems

COMPUTATIONAL METHODS FOR PSYCHOLOGICAL SCIENCE

A PSYCHOLOGIST'S GUIDE TO MACHINE LEARNING

ZURICH, MARCH 16-20, 2026



EMOTION REGULATION STRATEGIES

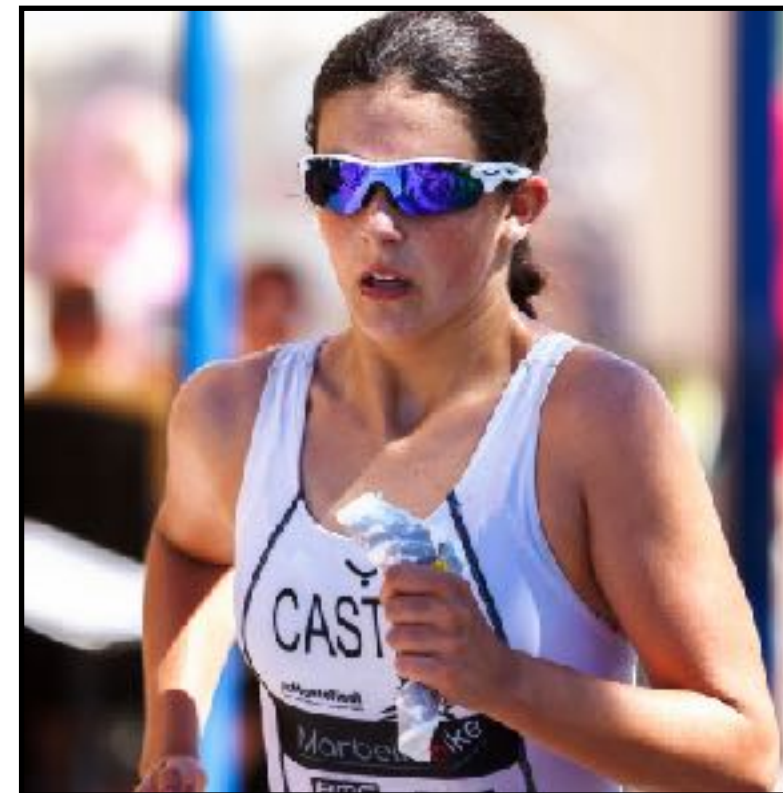


HATE & COUNTER SPEECH ON THE INTERNET

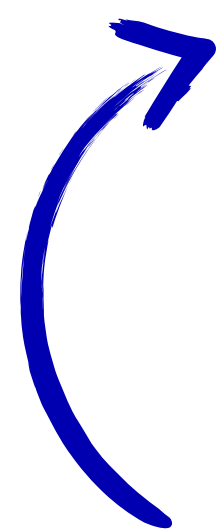
**A COMPUTATIONAL METHOD TO
REVEAL PSYCHOLOGICAL
CONSTRUCTS FROM TEXT DATA**
(HERDERICH ET AL., 2024)

INSPIRED BY A SIMPLE QUESTION: WHAT ER STRATEGIES ARE THERE?

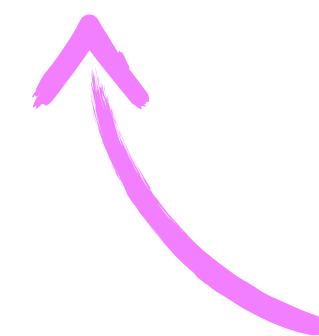
up-regulate
positive emotions



regulate others



down-regulate
negative emotions



regulate self

LOOKING AT THE MOST POPULAR ASSESSMENT TOOLS: NOT MANY.

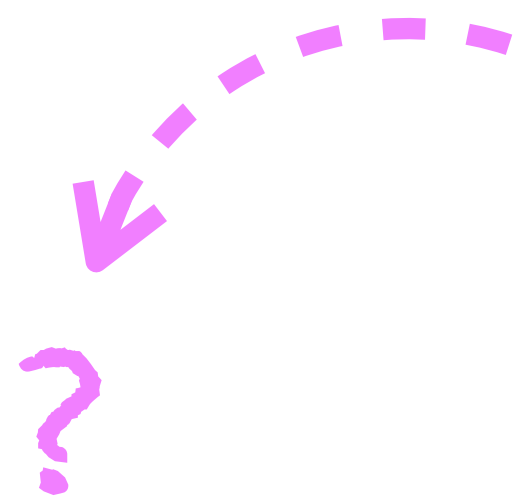
EMOTION REGULATION QUESTIONNAIRE (ERQ)

Rate on a scale from 1 to 7:

- When I want to feel more positive emotion, I change what I'm thinking about.
- I keep my emotions to myself.
- When I want to feel less negative emotion, I change what I'm thinking about.
- When I am feeling positive emotions, I am careful not to express them.
- ...

reappraisal

suppression



THERE ARE LIKELY MORE ER STRATEGIES THAN TWO.



reappraisal
suppression



reappraisal
suppression



reappraisal
suppression

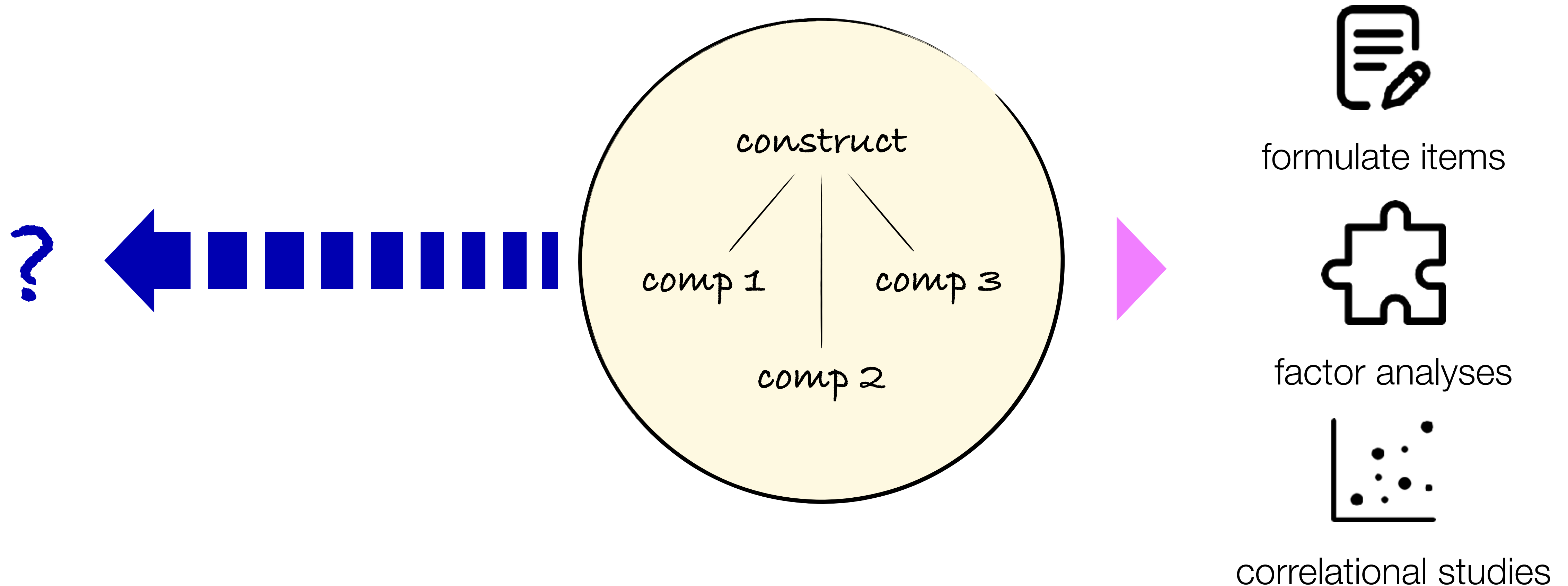
situation modification

distraction

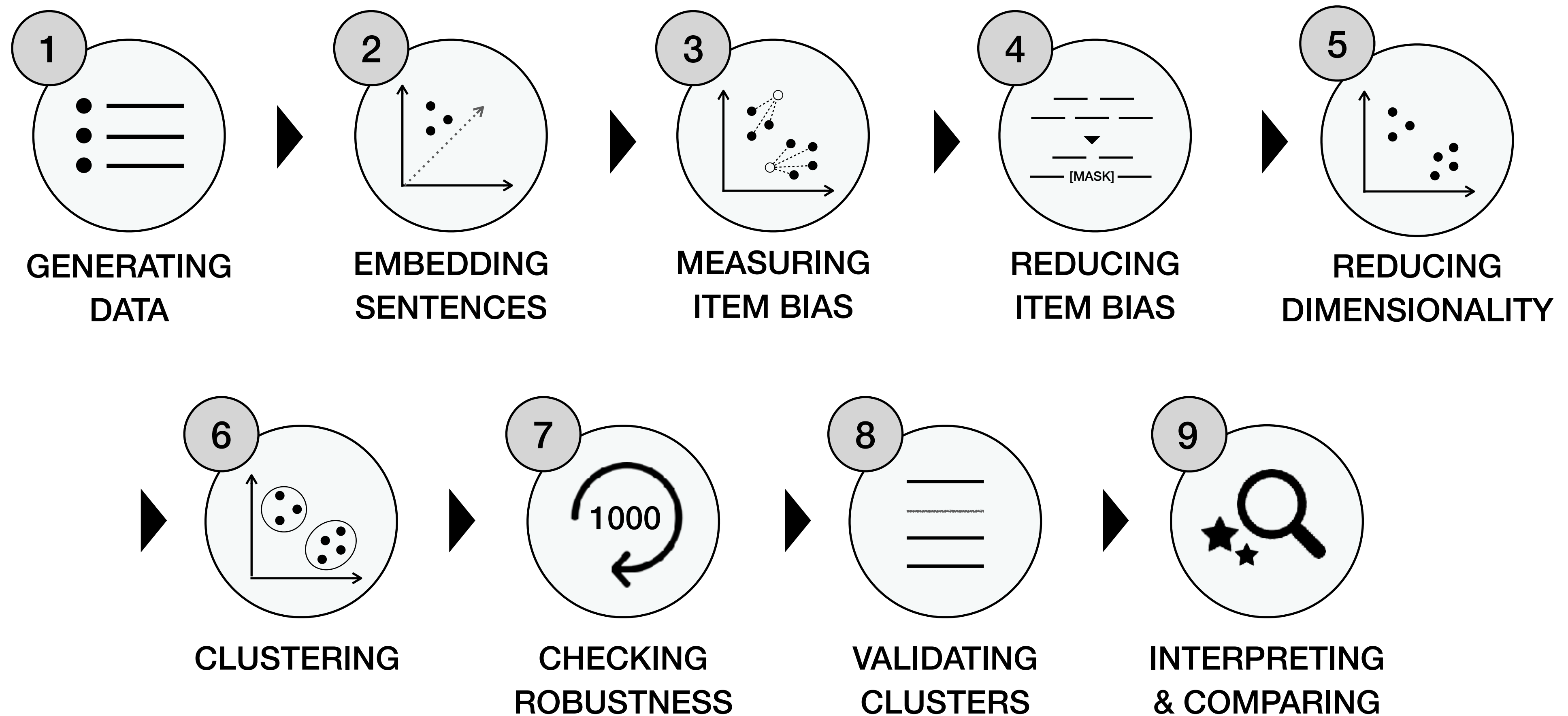
social support

variation happens here

FORMULATING CONSTRUCTS THROUGH THEORY ONLY MIGHT LEAVE GAPS.



THE CONSTRUCT MINING PIPELINE



1. WE COLLECTED ER STRATEGIES USING OPEN ENDED QUESTIONS.

Please vividly imagine the following situation:

You find out that some members of your social sports team have been saying that you are not a very good player.

Please list all ideas for changing your own feelings for the better in the short or the long run that you can think of.

THE EMOTION REGULATION STRATEGY INVENTIVENESS TASK

1. I seek help with the trainer.
2. I confront the team members.
3. I tell myself fun is most important.
4. I quit the team.
5. ...

2. WE EMBEDDED THE STRATEGIES IN A SEMANTIC SPACE.

n sentences

- ▶ This is
- ▶ A list of
- ▶ Sentences
- ▶ ...

A PRETRAINED
TRANSFORMER
ENCODER

n vectors of length d
(or n x d matrix)

- ▶ [18723, 92, 237, ...]
- ▶ [94, 8234, 9992, ...]
- ▶ [8734124, 23, 7, ...]
- ▶ ...

TRAINING

Step 1:

?

Paris is the [MASK] of France.

Step 2:

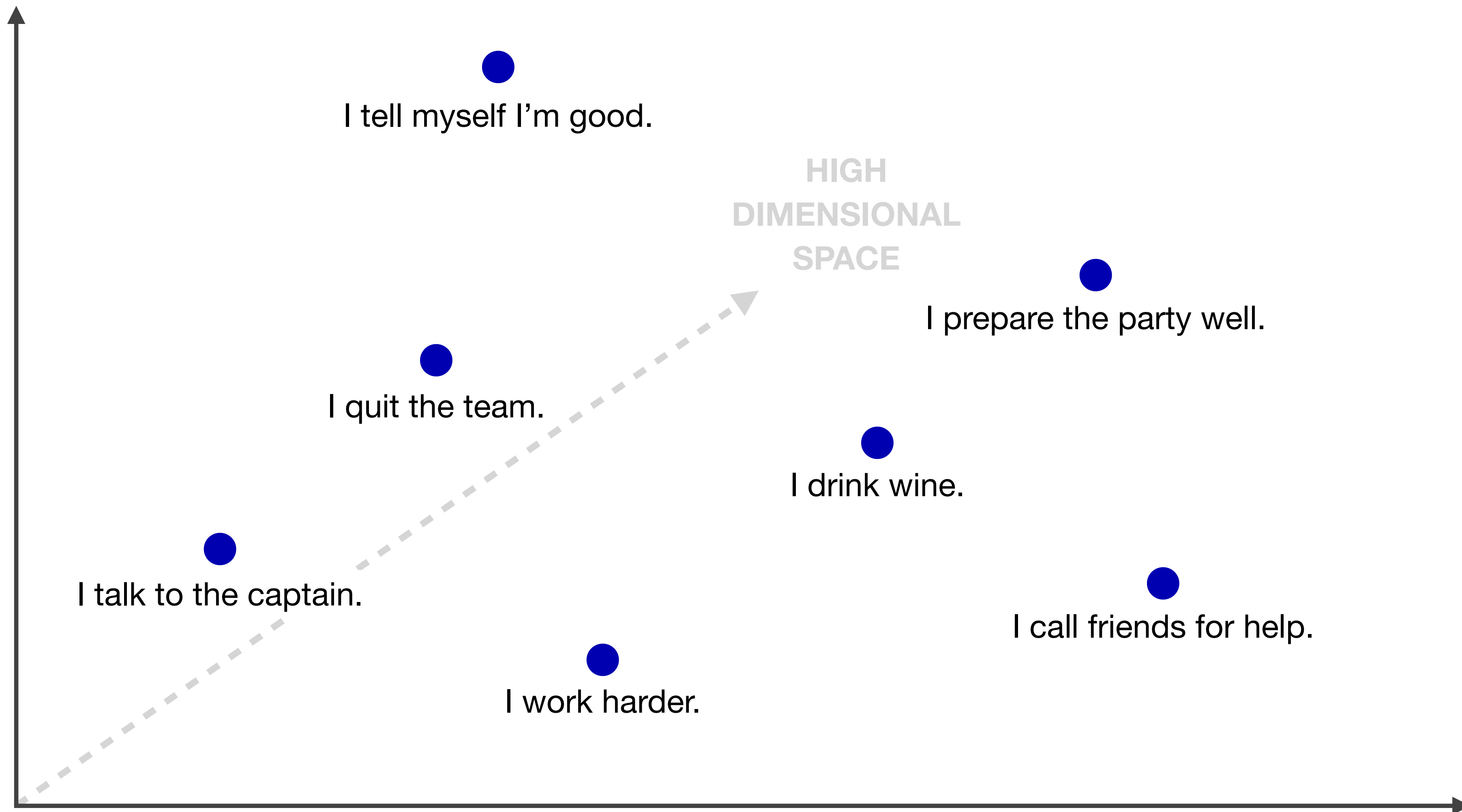
I love espresso.

I love iced latte.

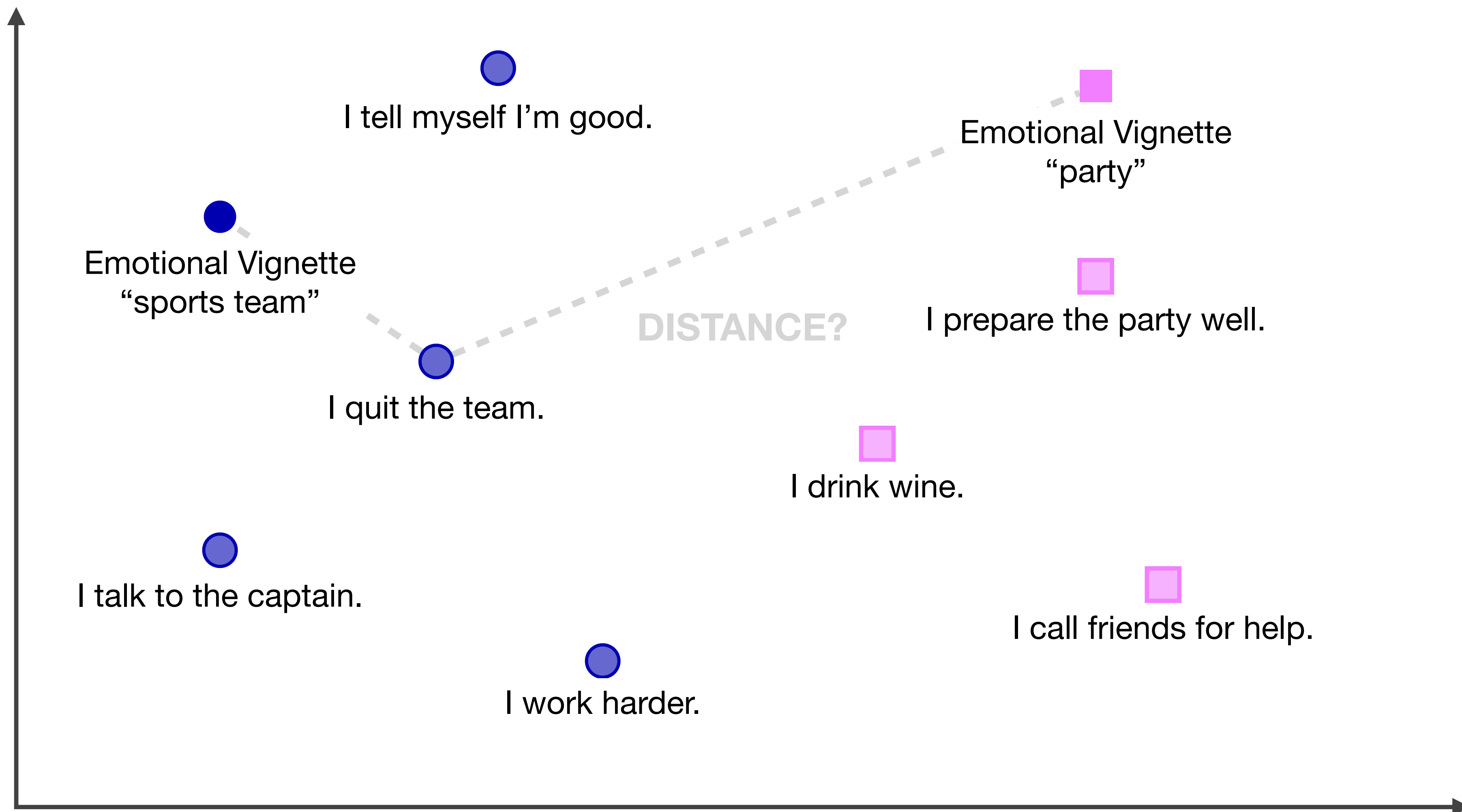
similarity

4.7/5

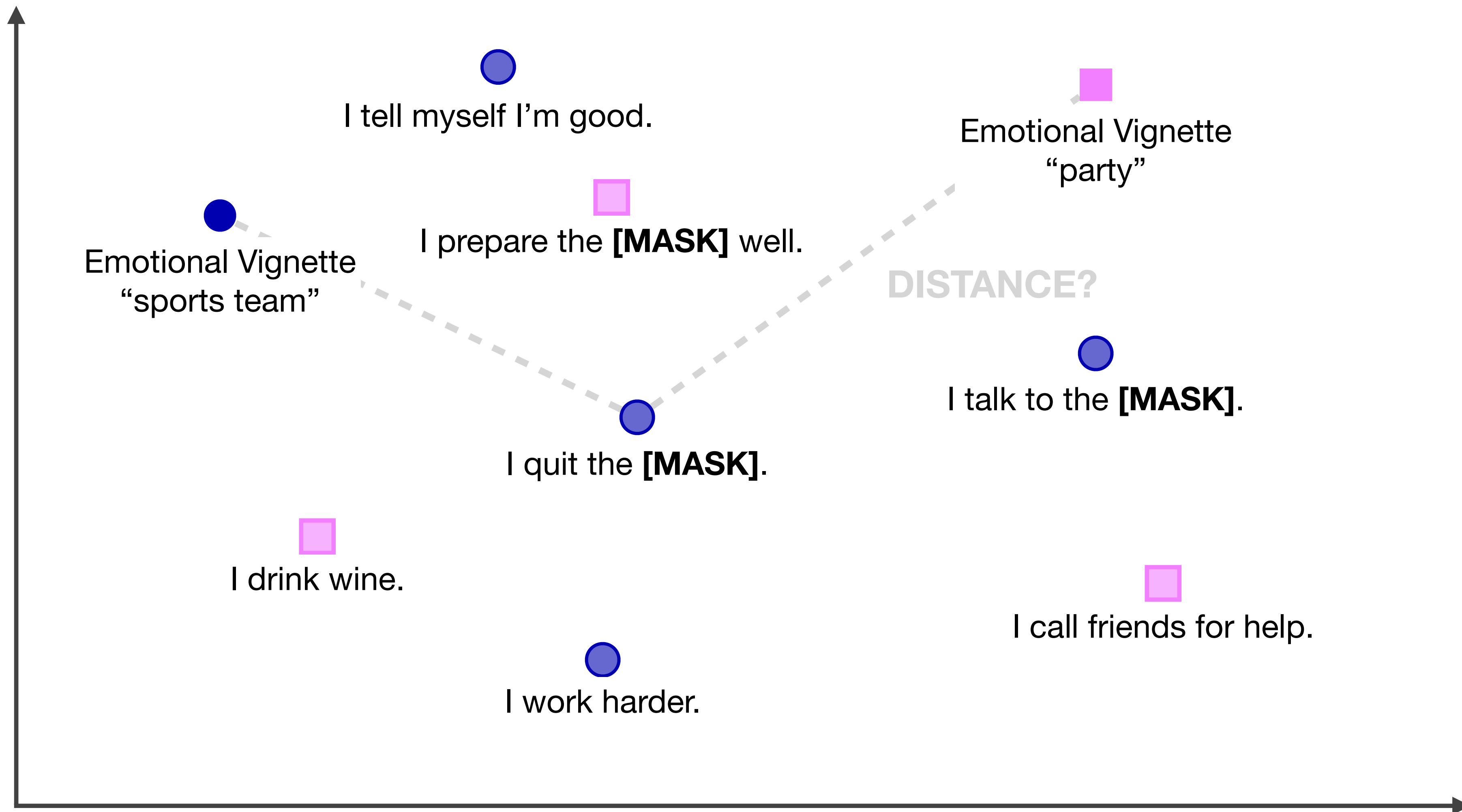
2. WE EMBEDDED THE STRATEGIES IN A SEMANTIC SPACE.



3. WE MEASURED THE INFLUENCE OF UNRELATED CONCEPTS IN THIS SPACE.



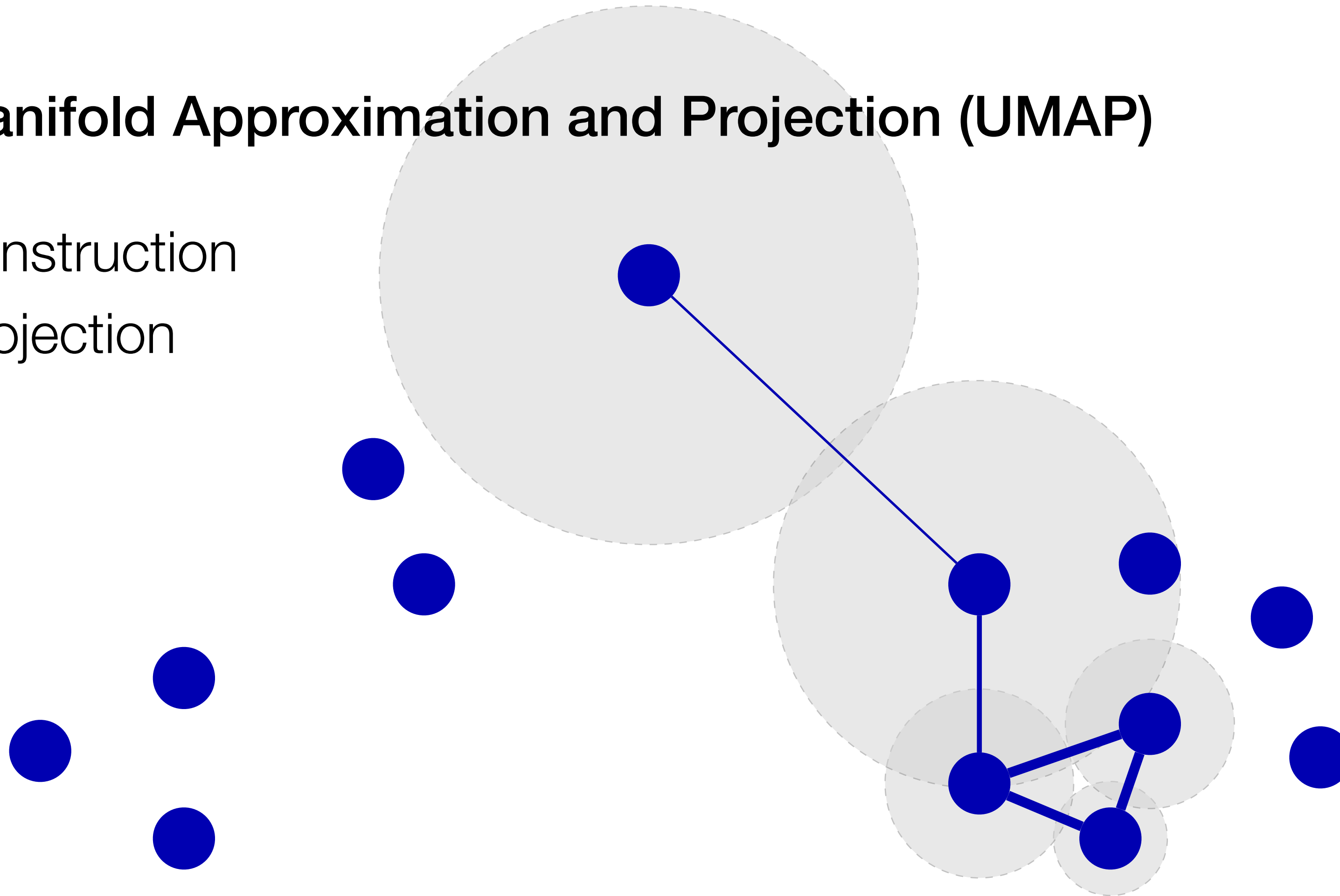
4. WE REMOVED UNRELATED CONCEPTS FROM THE EMBEDDING SPACE.



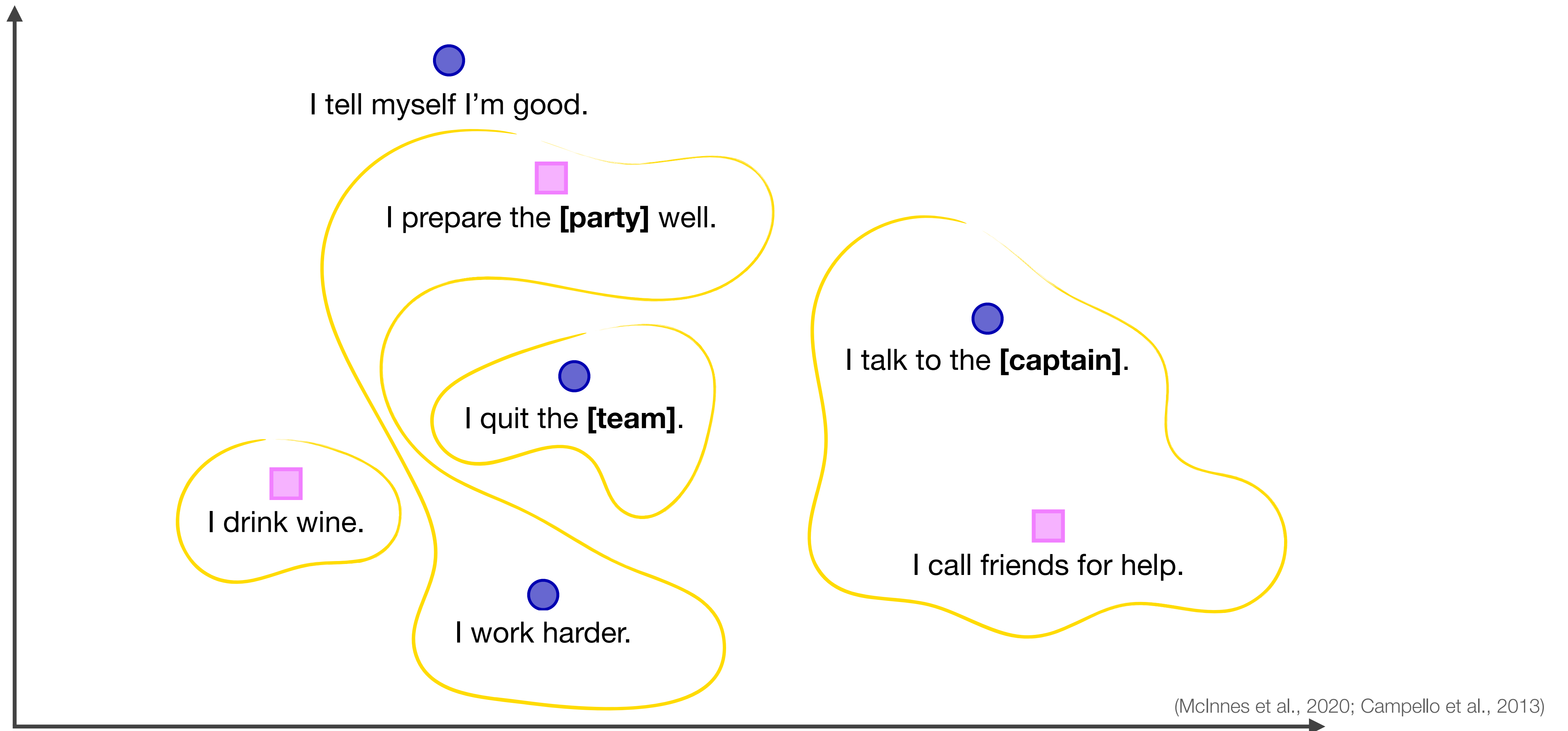
5. WE REDUCED THE DIMENSIONALITY OF THE EMBEDDING SPACE.

Uniform Manifold Approximation and Projection (UMAP)

1. Graph construction
2. Graph projection

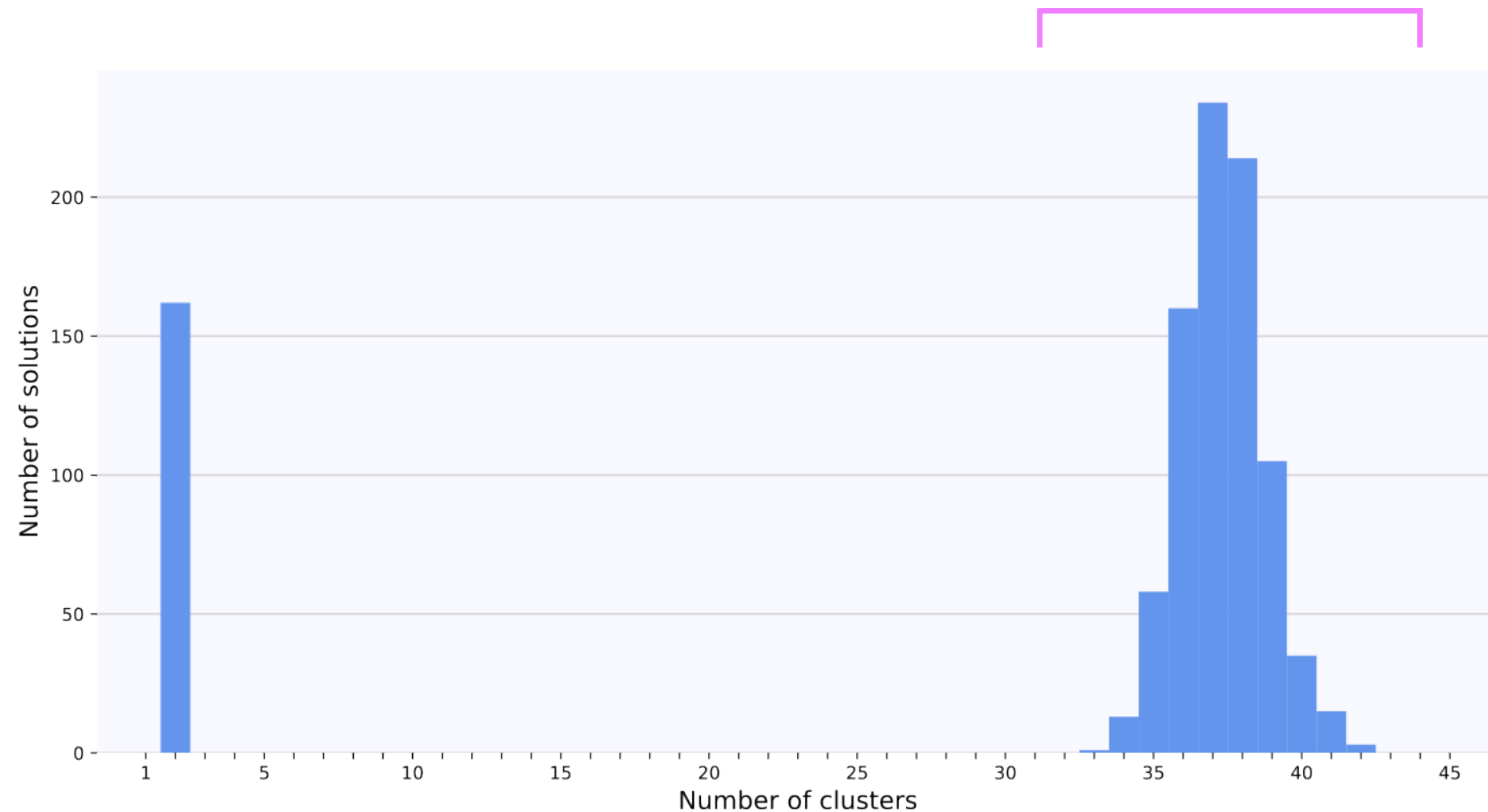


6. WE CLUSTERED ON THE EMBEDDINGS TO IDENTIFY CANDIDATE STRATEGIES.

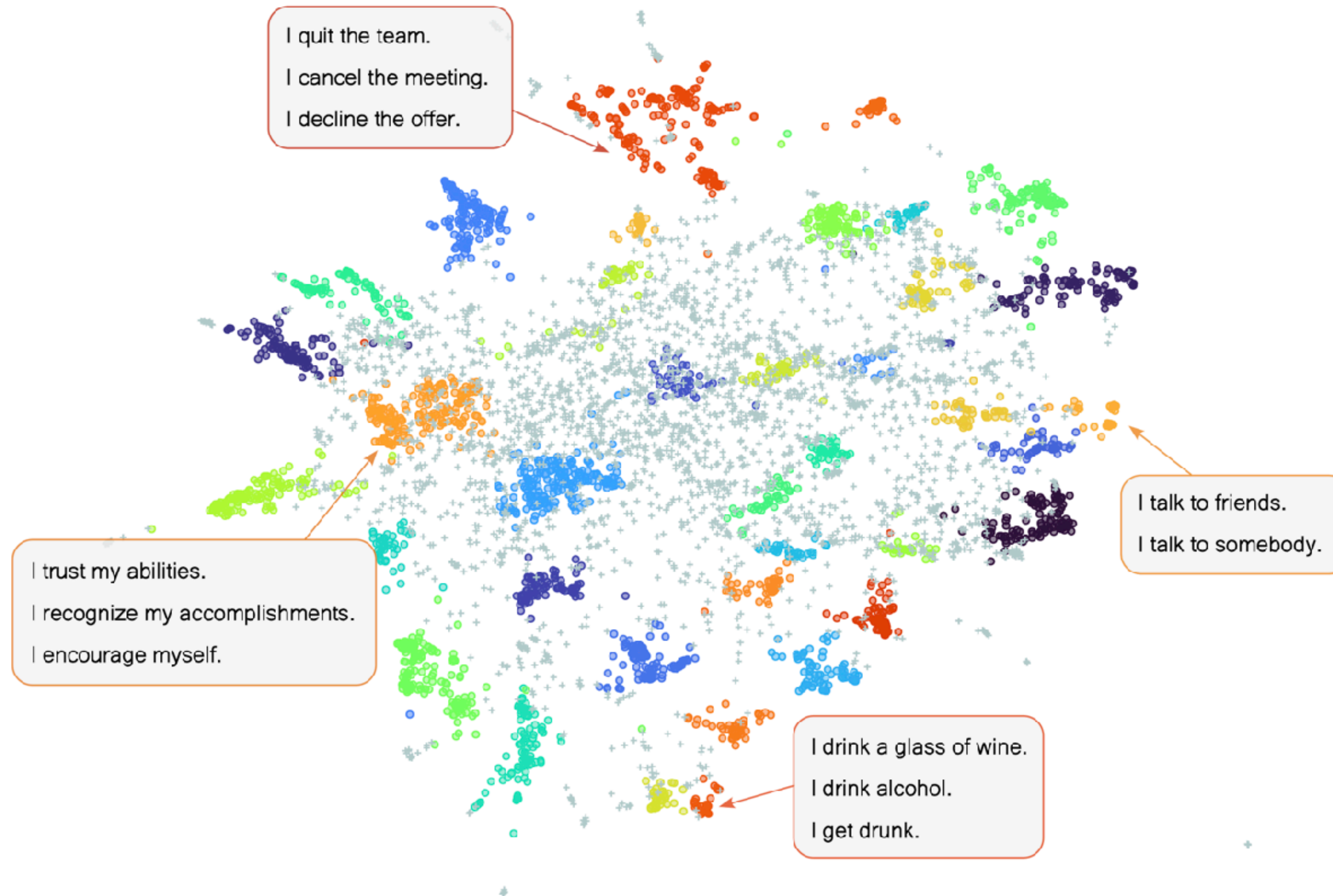


7. WE CONDUCT A ROBUSTNESS CHECK TO AVOID RESULTS BY CHANCE.

Adjusted Rand Indices = [0.65, 0.88]



TADA 🎉 THE FINAL CLUSTERING!



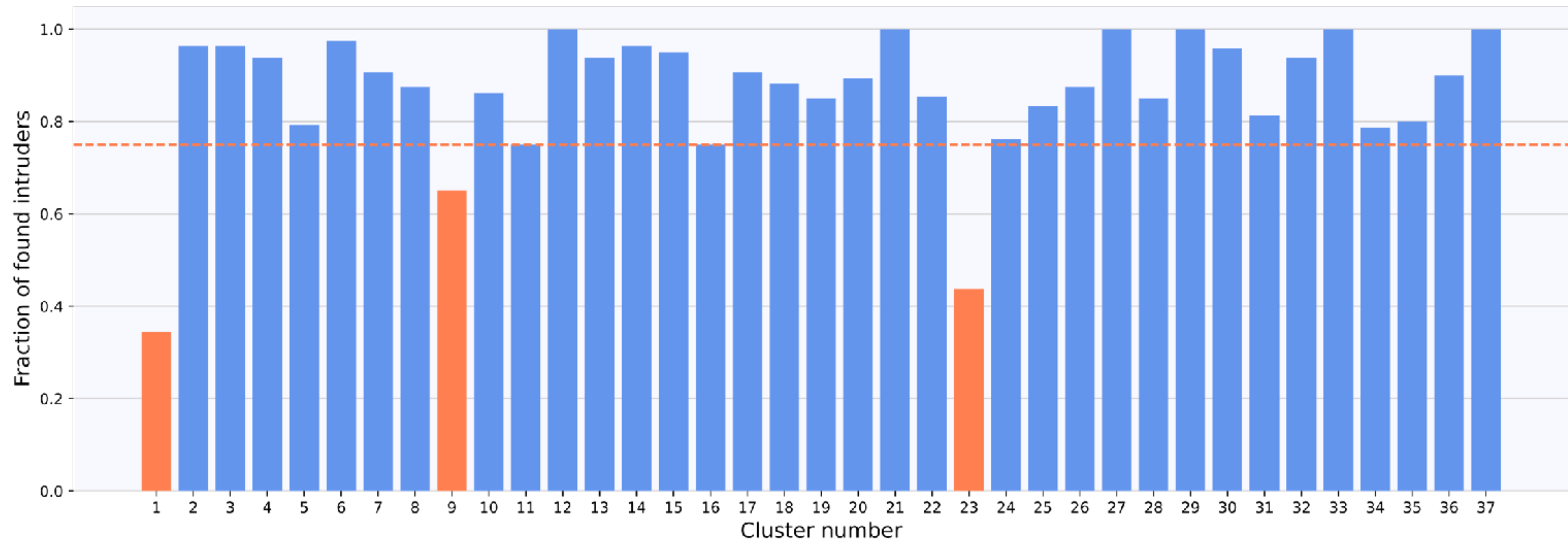
8. WE ESTABLISH ANOTHER LAYER OF VALIDATION WITH HUMANS IN THE LOOP.

SENTENCE INTRUSION TASK

Which of the following sentences does not belong to the others?

- ☐ I call a friend to talk about it.
- ☐ I tell my friend how it went.
- ☐ I just go home and cry.
- ☐ I turn to my friend for advice.

8. WE ESTABLISH ANOTHER LAYER OF VALIDATION WITH HUMANS IN THE LOOP.



9. WE INTERPRETED THE REMAINING CLUSTERS AS STRATEGY CLASSES.

Substance Use
Suppression
Distraction
Emotional Expression
Social Support
Reappraisal
Relaxation Techniques
Therapy

well-known classes

Situation Modification
Instrumental Support
Withdrawal
Eating & Drinking

*classes known, but
not sufficiently
discussed*

Information Seeking
Self-Enhancement

*unknown
classes*

Rumination
Self-harm
...

*known, but
unidentified
classes*

THIS IS A GREAT STUDY, BECAUSE

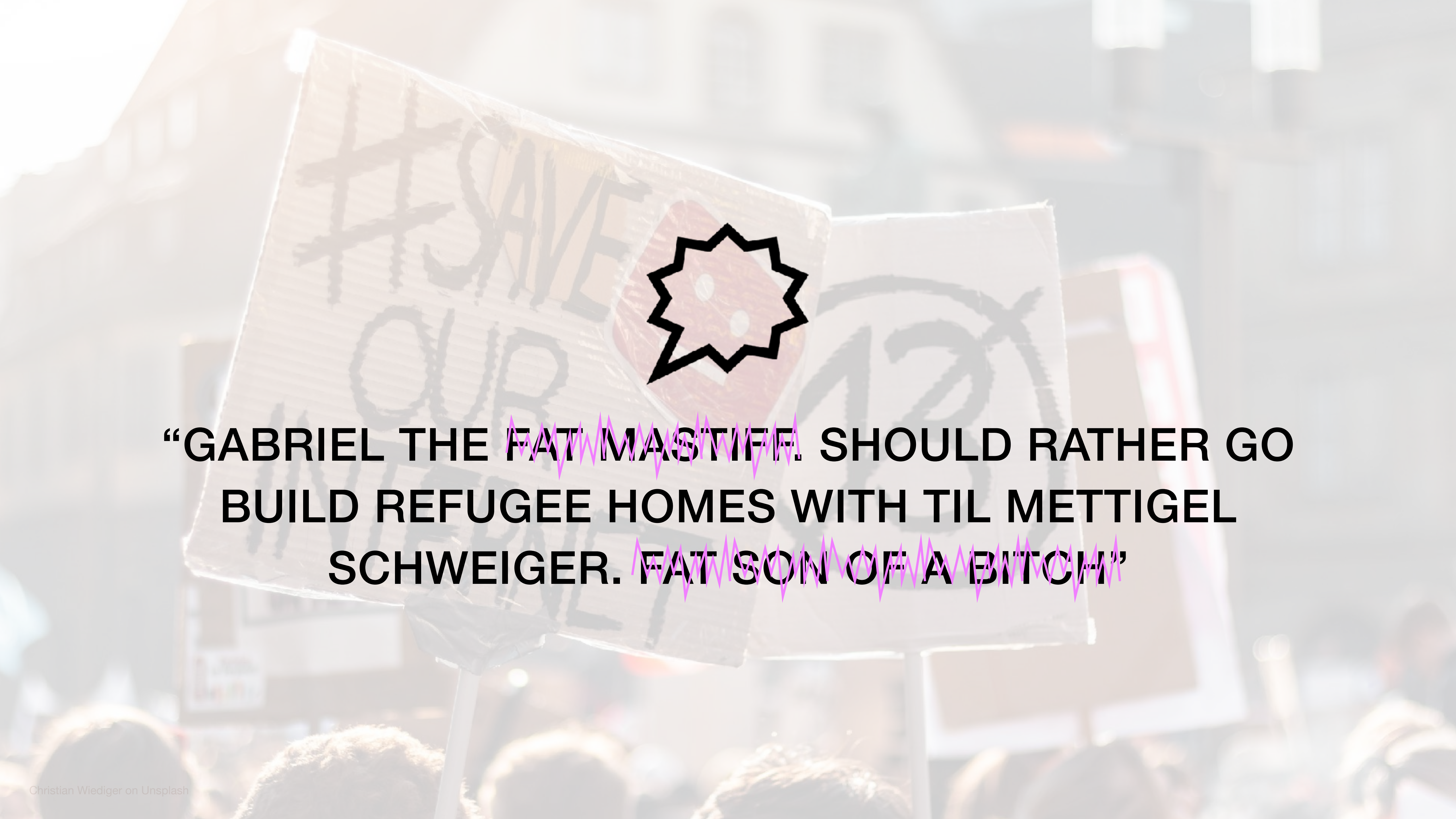
- ▶ We circumvent rating scales with people interpreting the construct of interest directly.
- ▶ We reduce researchers' degrees of freedom with a computer assisted approach.
- ▶ Findings are replicable, extensible, and scalable.

BUT IT ALSO HAS LIMITATIONS.

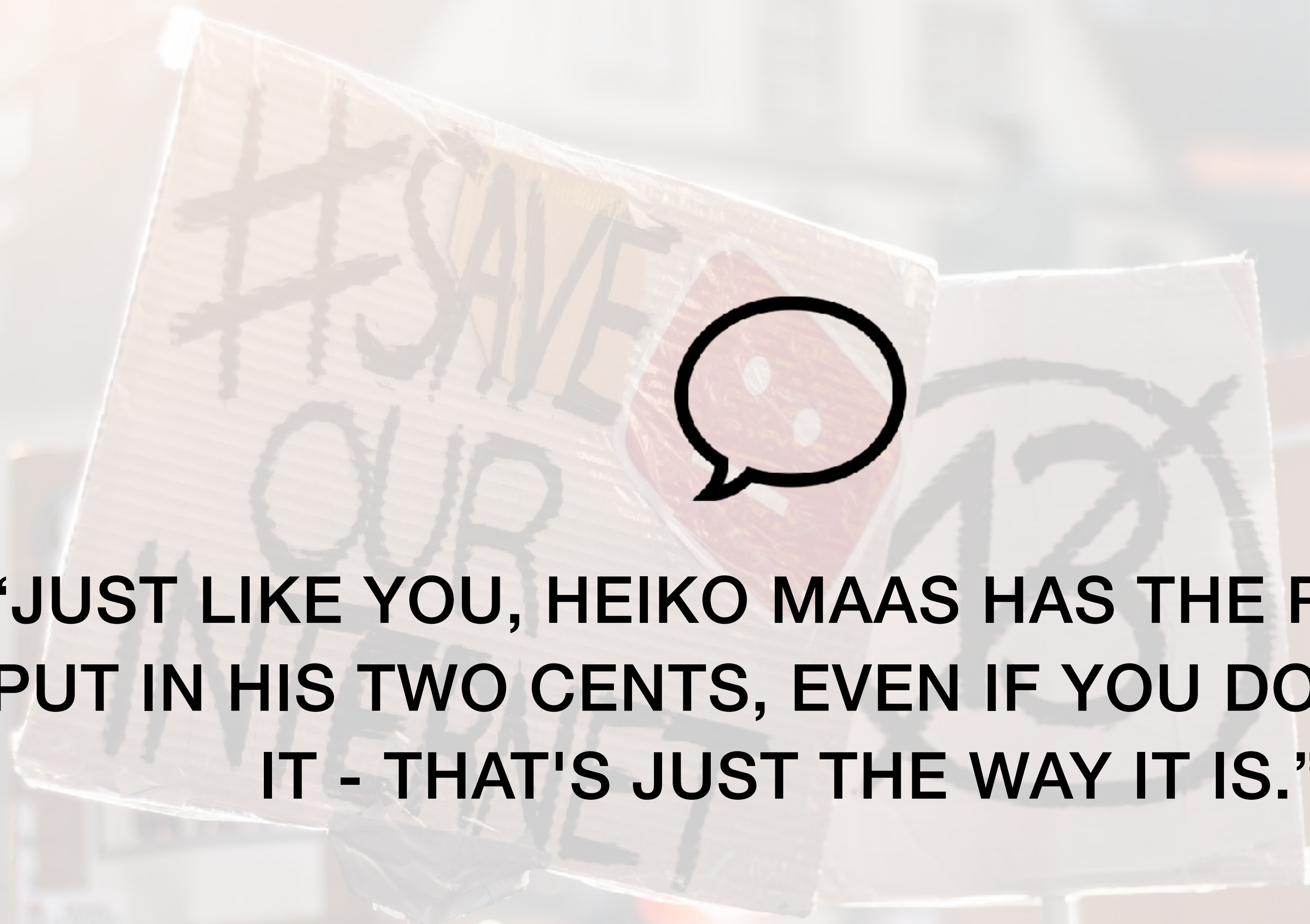
- ▶ There are still considerable degrees of freedom.
- ▶ The method works well for some constructs, but not others.
- ▶ The method works well for identifying classes, not assigning them.
- ▶ The method needs to be tested further.

COLLECTIVE MODERATION OF HATE, TOXICITY AND EXTREMITY IN ONLINE DISCUSSIONS

(LASSER, HERDERICH ET AL., 2025)

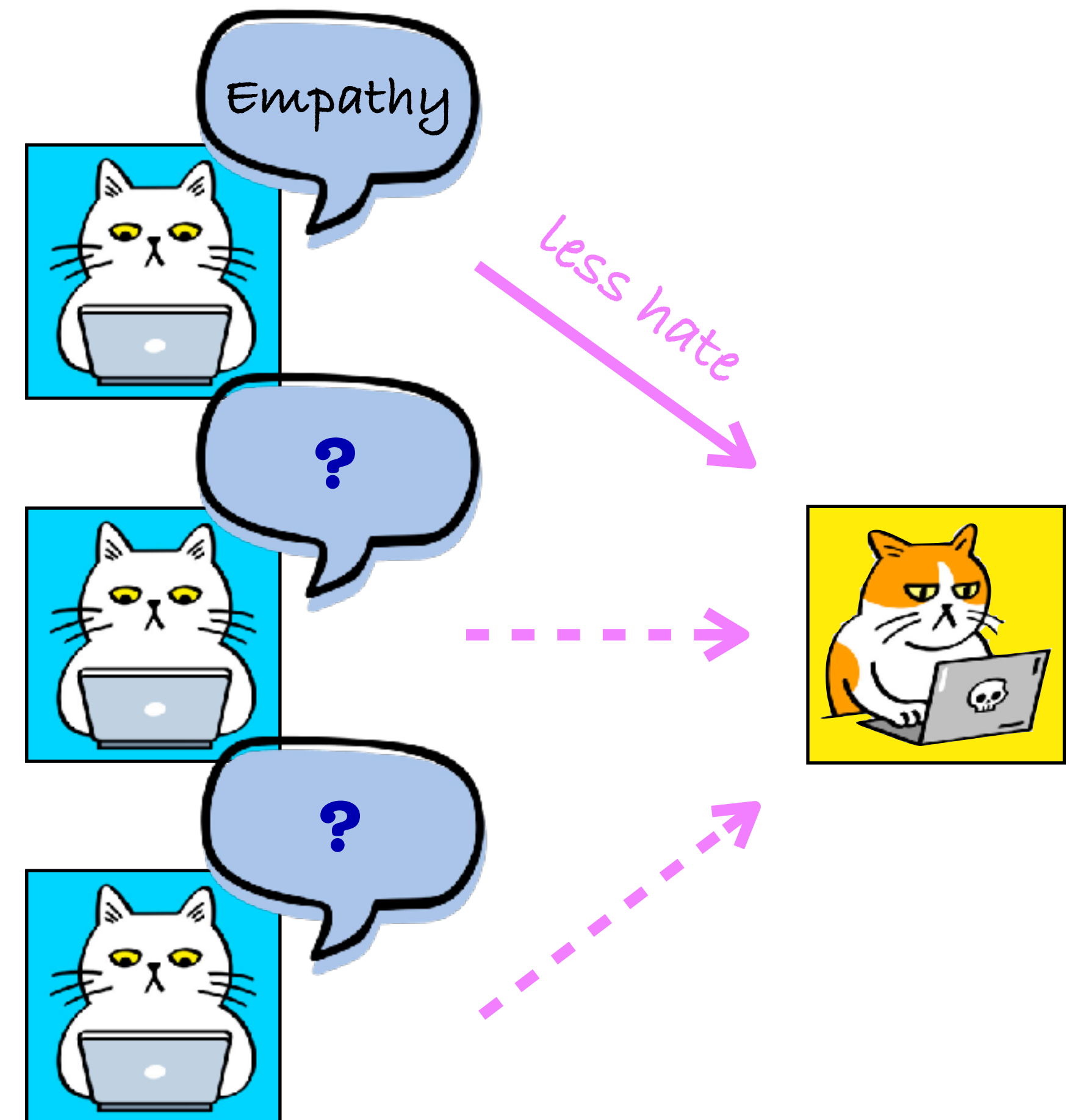
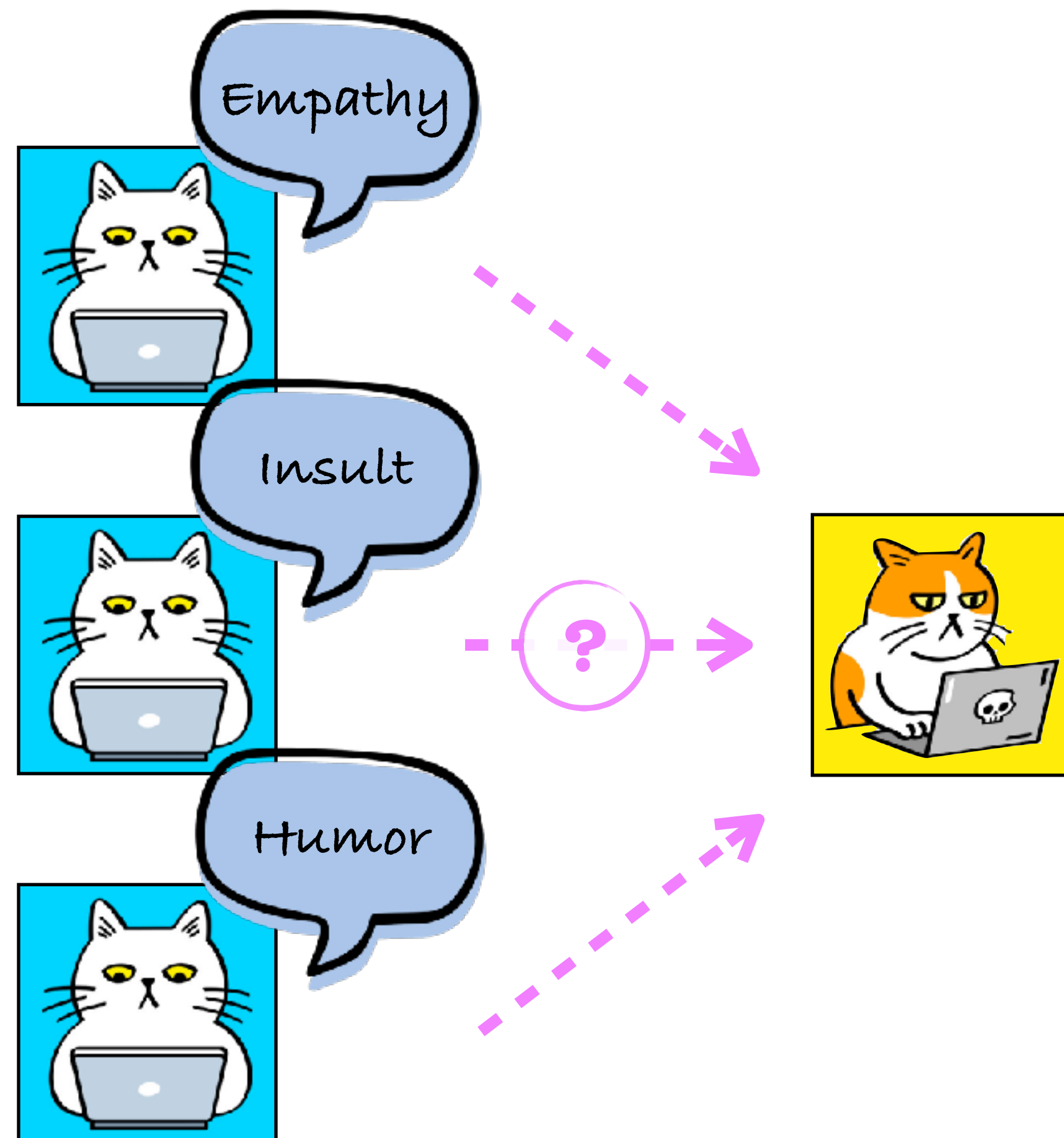


**“GABRIEL THE ~~FAT MASTIFF~~ SHOULD RATHER GO
BUILD REFUGEE HOMES WITH TIL METTIGEL
SCHWEIGER. ~~FAT SON OF A BITCH~~”**

A protest sign made of cardboard with the text "#SAVE OUR FREEDOM" written on it. A speech bubble icon is overlaid on the sign. The background is a blurred crowd of people at a protest.

**“JUST LIKE YOU, HEIKO MAAS HAS THE RIGHT TO
PUT IN HIS TWO CENTS, EVEN IF YOU DON’T LIKE
IT - THAT’S JUST THE WAY IT IS.”**

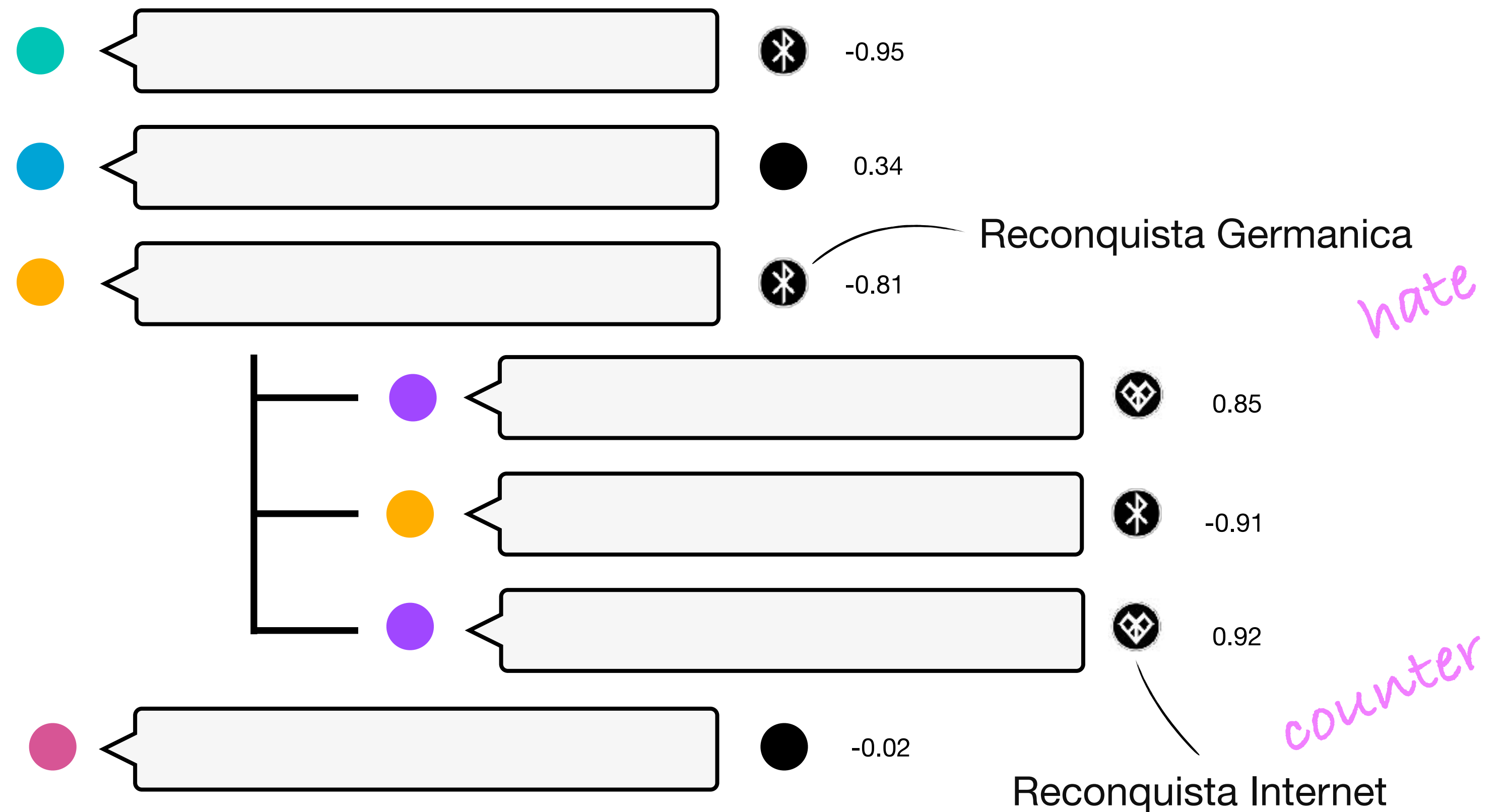
WE DON'T KNOW HOW COUNTER SPEECH PLAYS OUT AT SCALE.



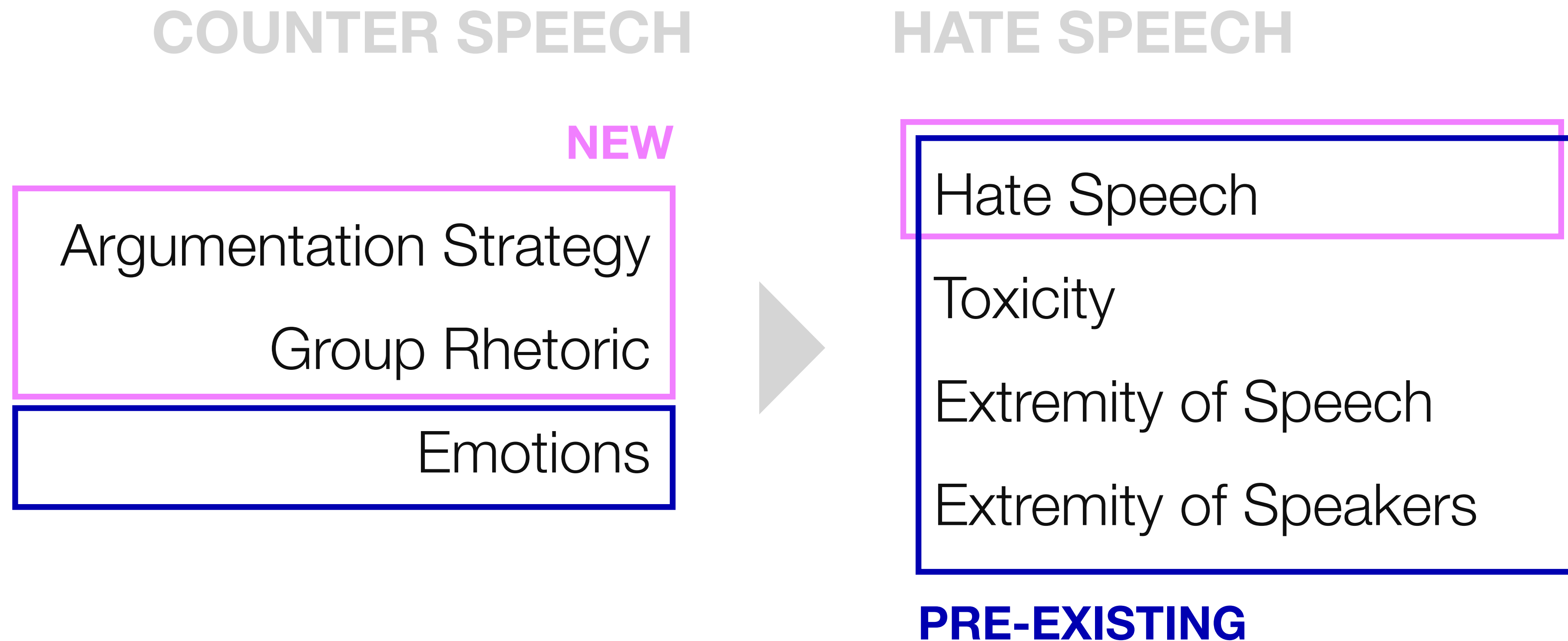
WE ANALYZED A LARGE DATASET OF GERMAN TWITTER CONVERSATIONS.



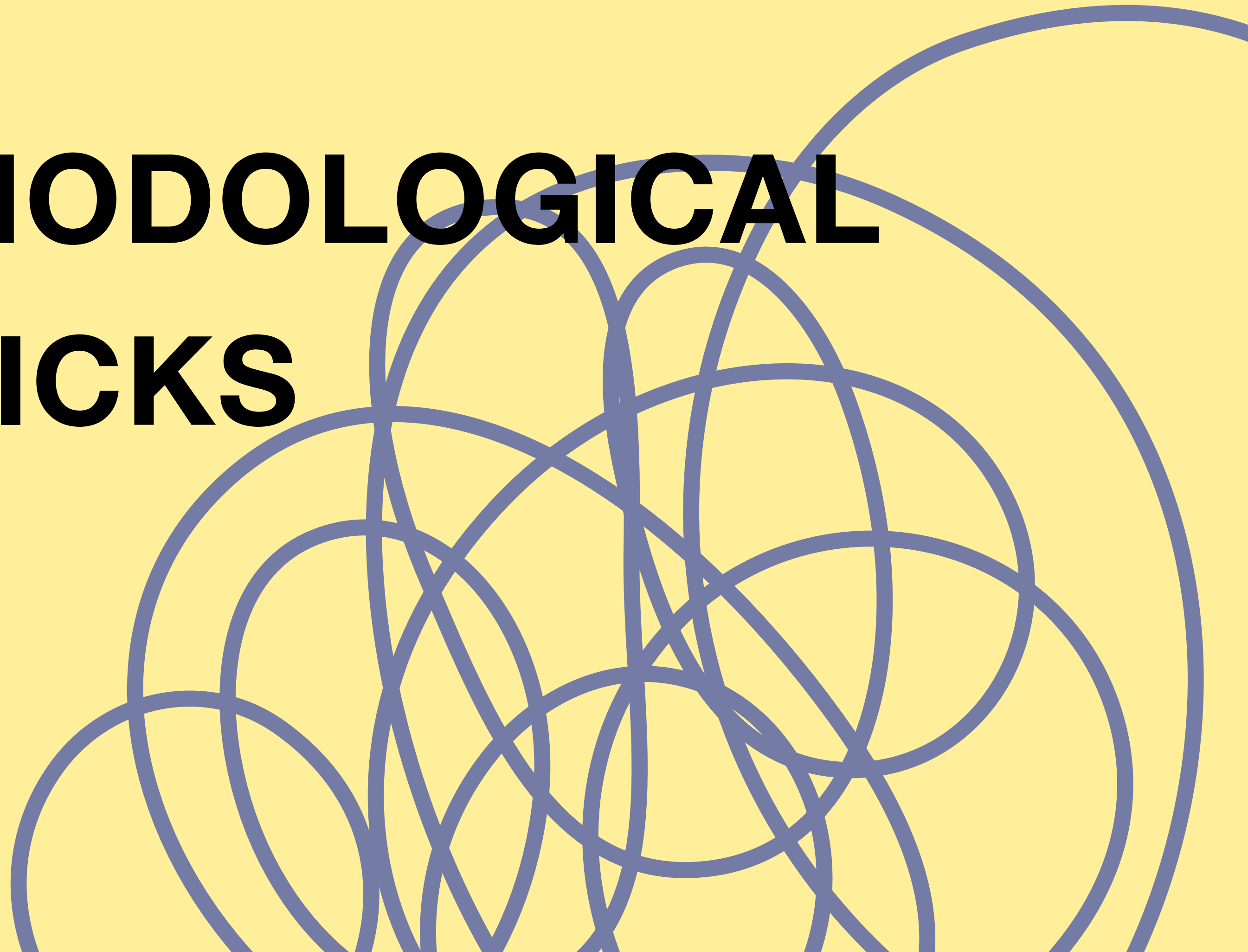
Years: 2015 - 2018
Discussions: 130,127
Tweets: 1,150,469
Users: 130,548



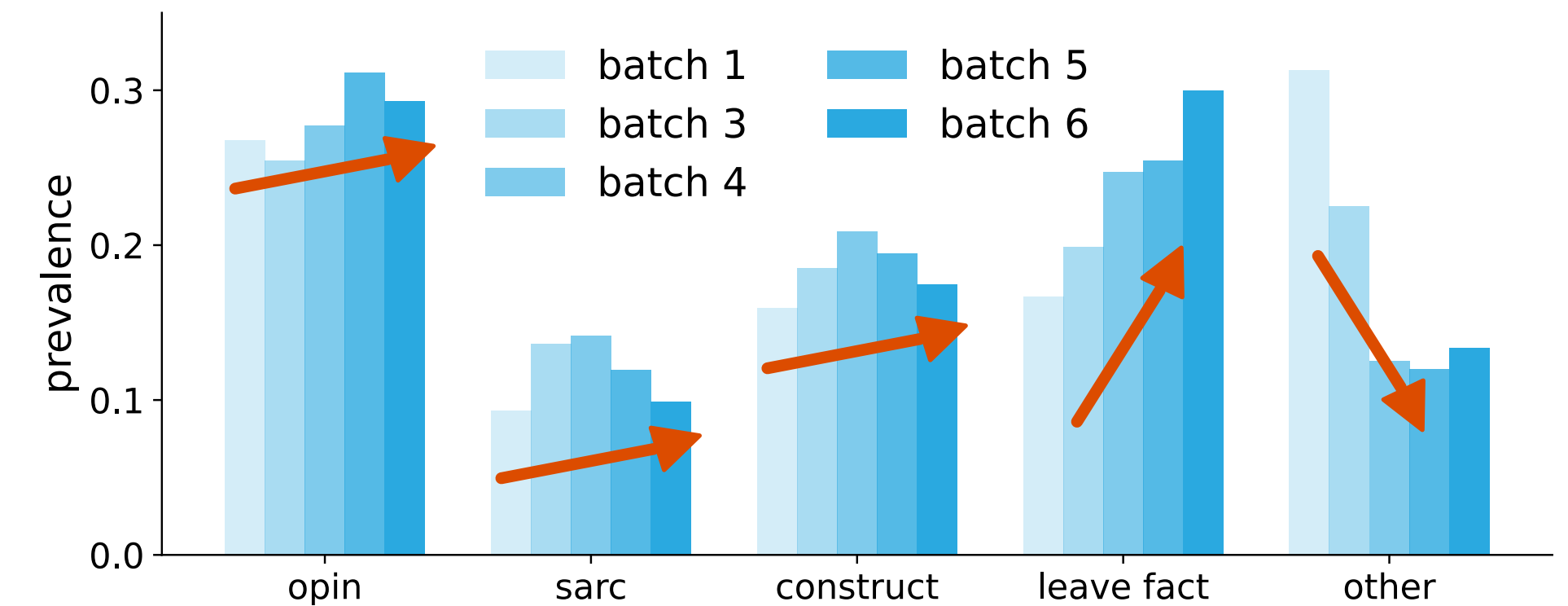
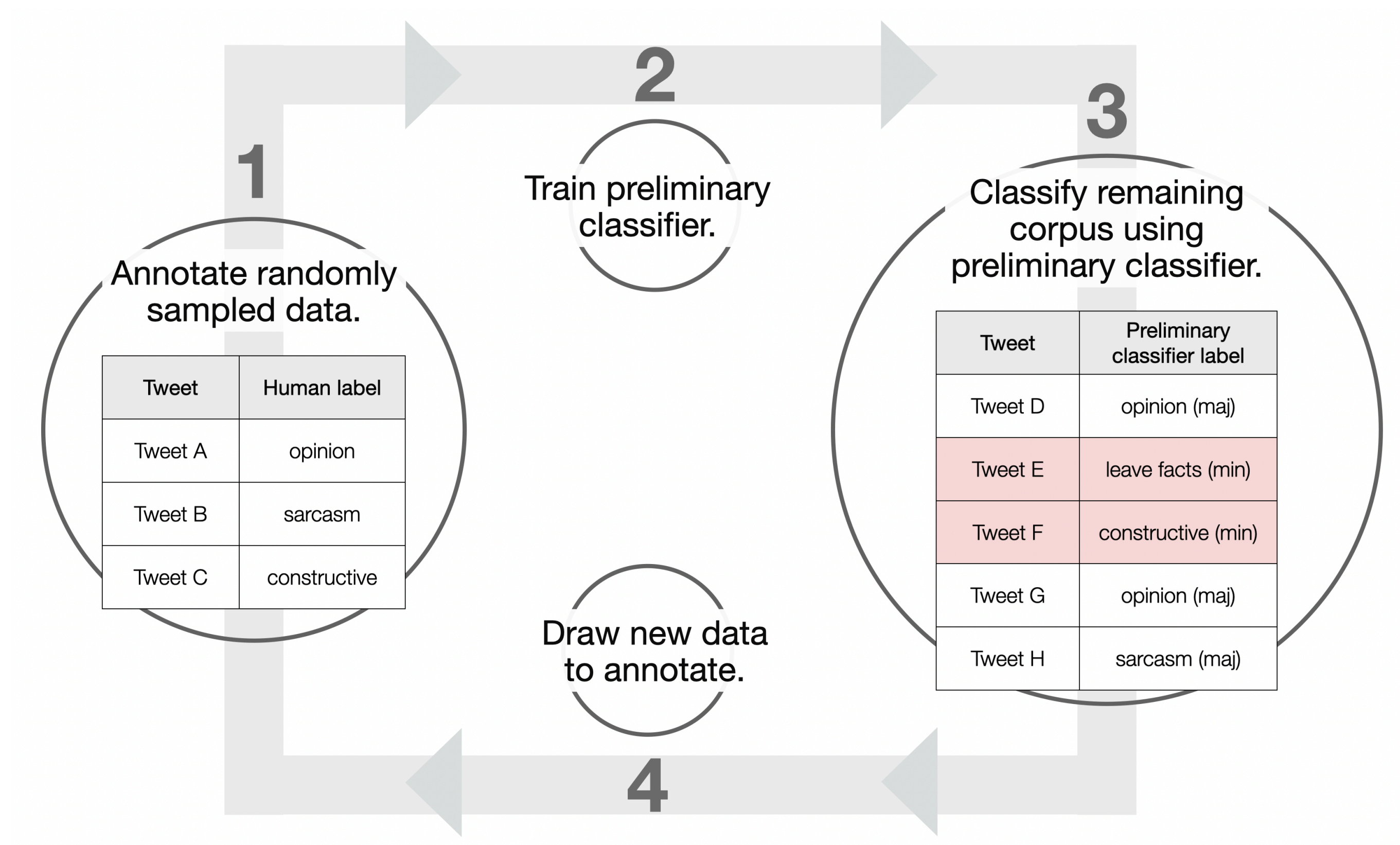
WE MEASURED HATE/COUNTER SPEECH WITH NEW AND EXISTING CLASSIFIERS.



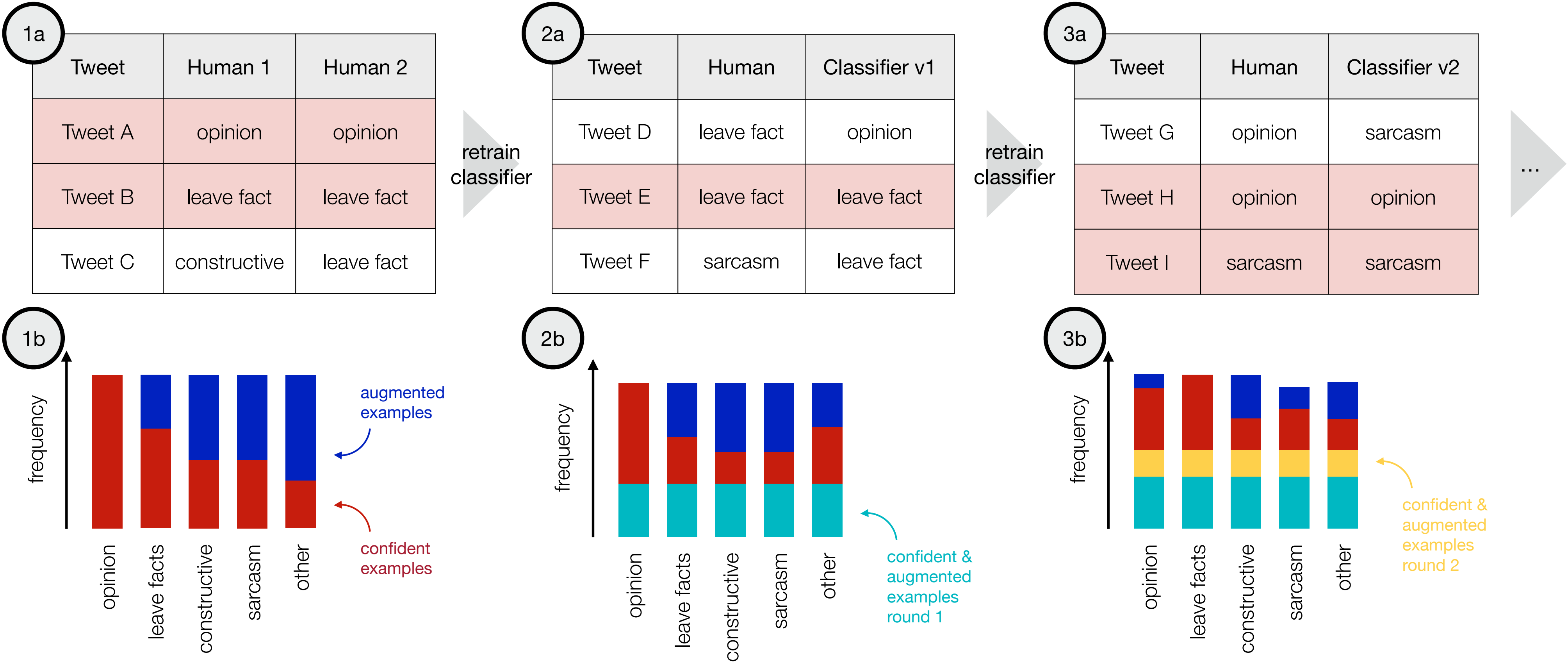
SOME METHODOLOGICAL TRICKS



WE BALANCED THE TRAINING DATA WITH ITERATIVE ANNOTATION.



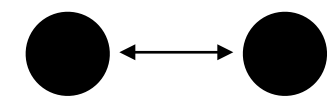
WE BOOSTED CLASSIFIER PERFORMANCE USING CONFIDENT EXAMPLES.



WE ANALYZED THE RELATIONSHIP ON THREE LEVELS OF AGGREGATION.

MICRO

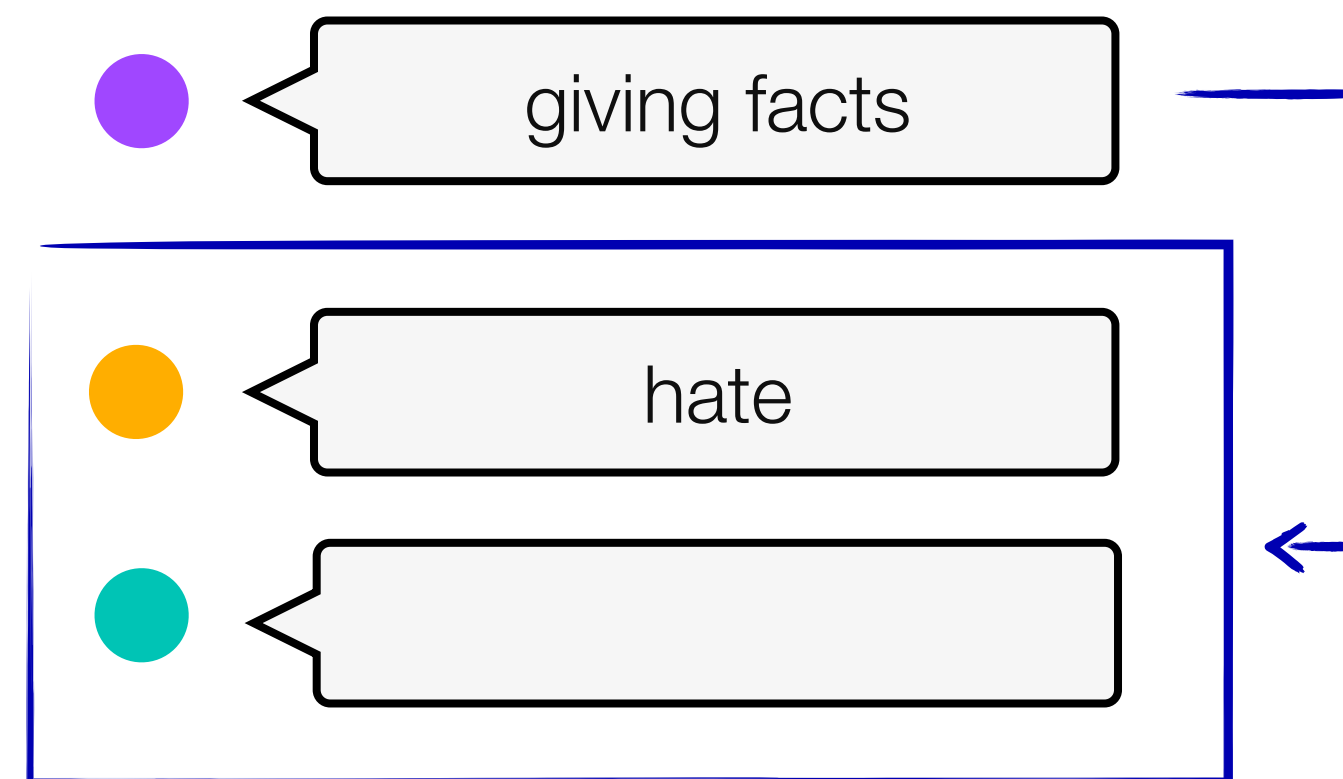
Individual Reply Pairs



MATCHING

MESO

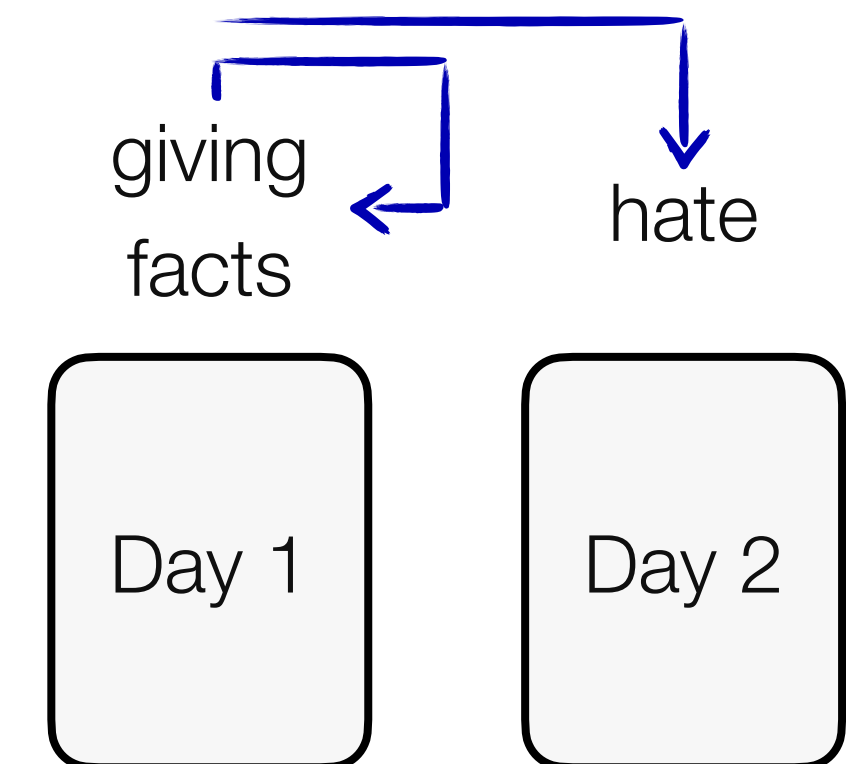
Discussions



ARDL

MACRO

Days



ARDL

THREE MAIN FINDINGS: OPINIONS - YES, GROUP RHETORIC - NO, EMOTIONS - NO.

REPLY PAIRS

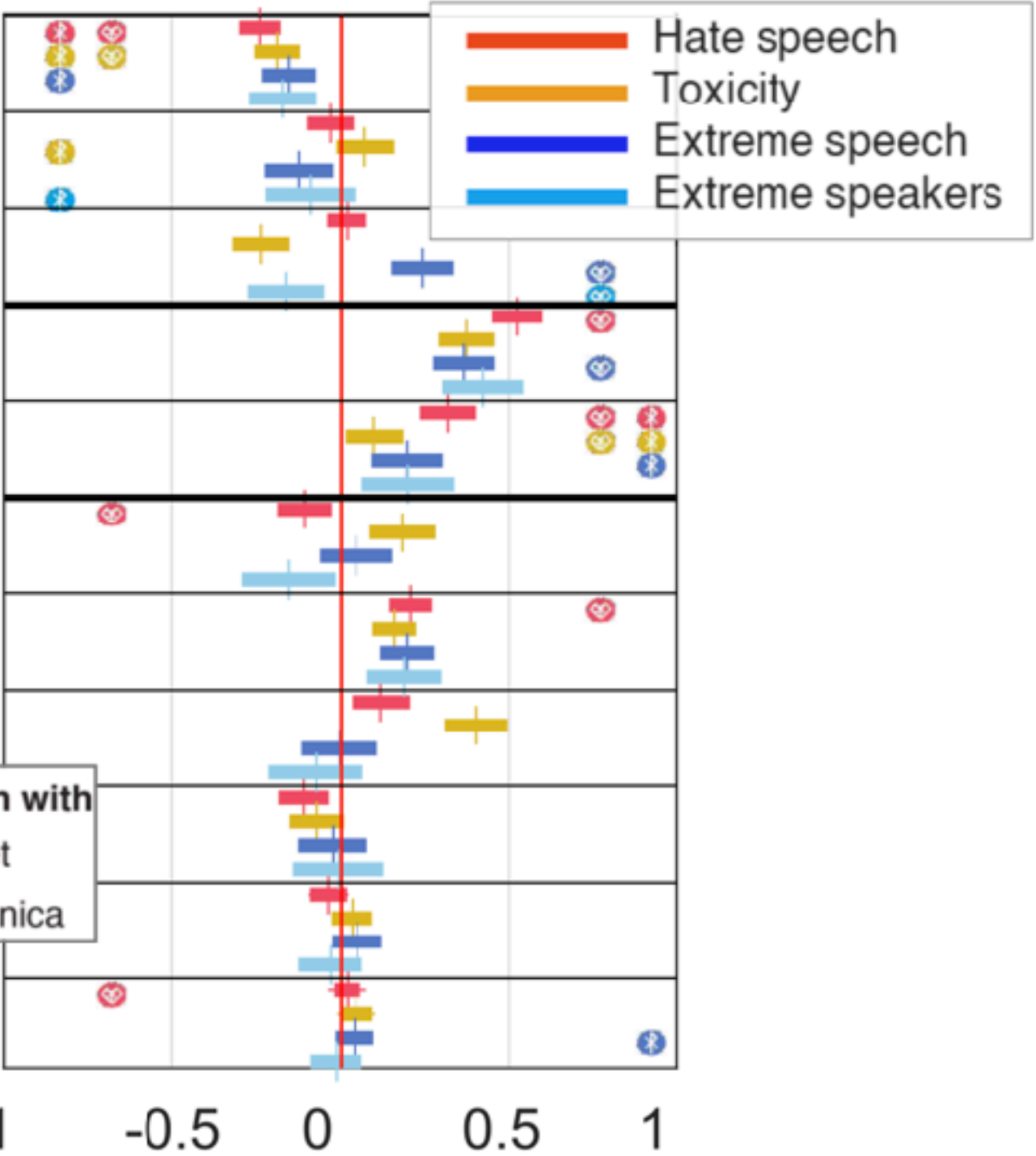
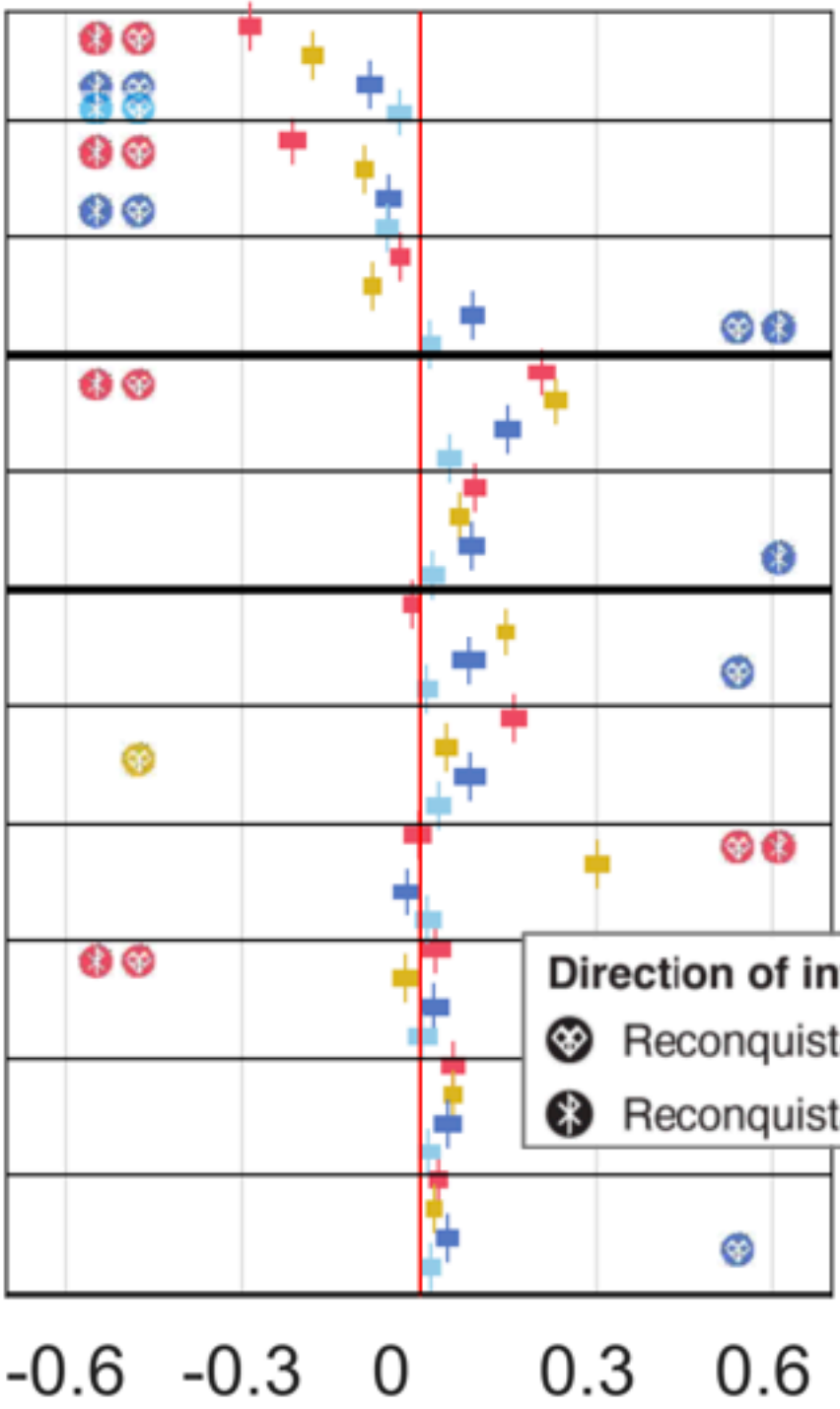
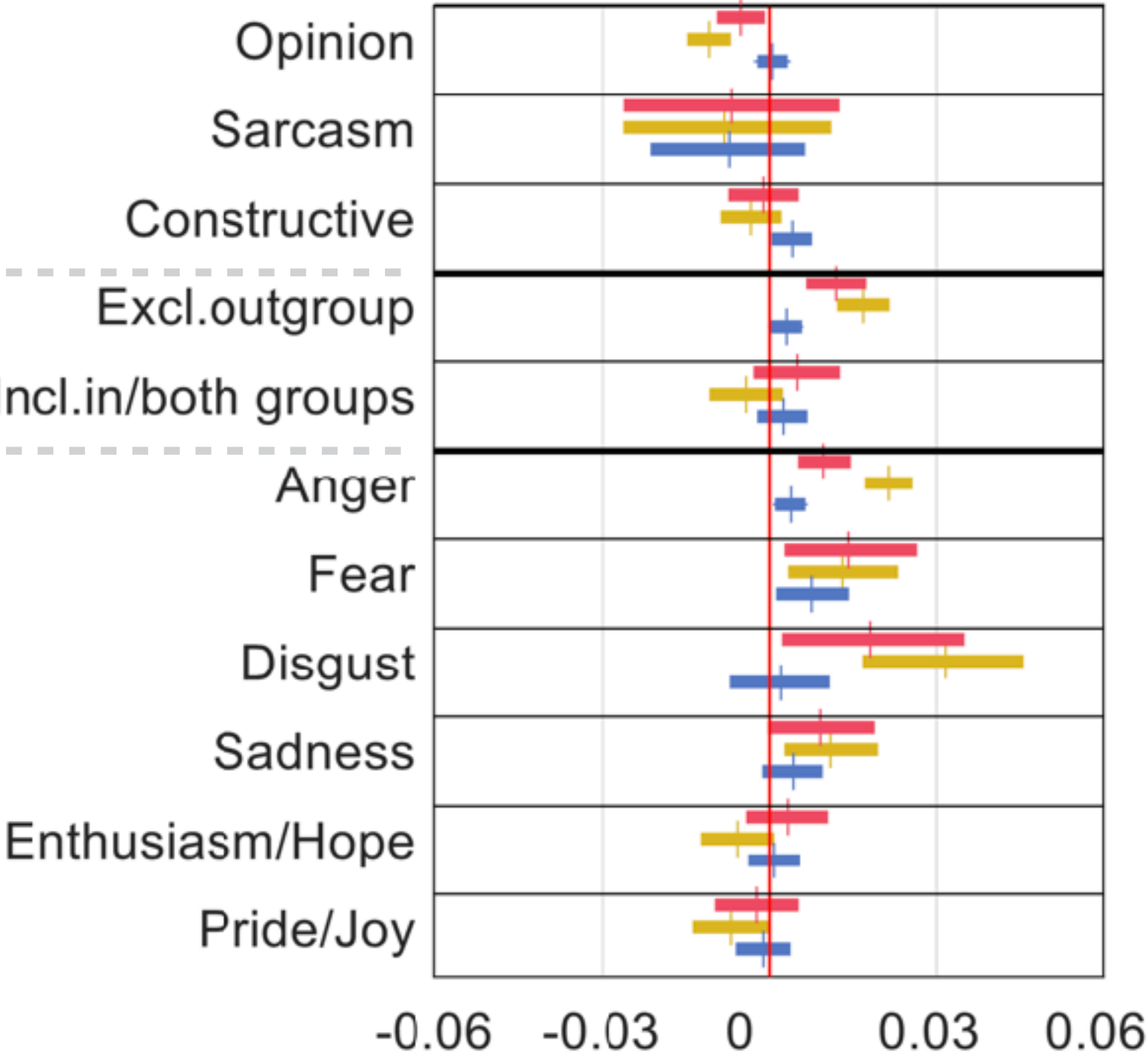
DISCUSSIONS

DAYS

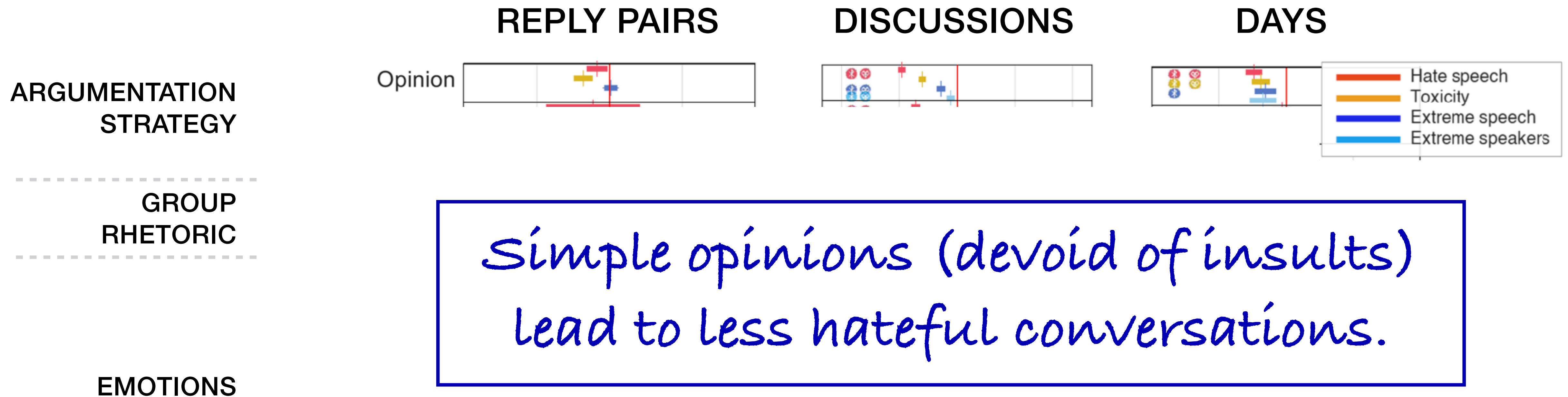
ARGUMENTATION
STRATEGY

GROUP
RHETORIC

EMOTIONS



THREE MAIN FINDINGS: OPINIONS - YES, GROUP RHETORIC - NO, EMOTIONS - NO.



THREE MAIN FINDINGS: OPINIONS - YES, GROUP RHETORIC - NO, EMOTIONS - NO.

REPLY PAIRS

DISCUSSIONS

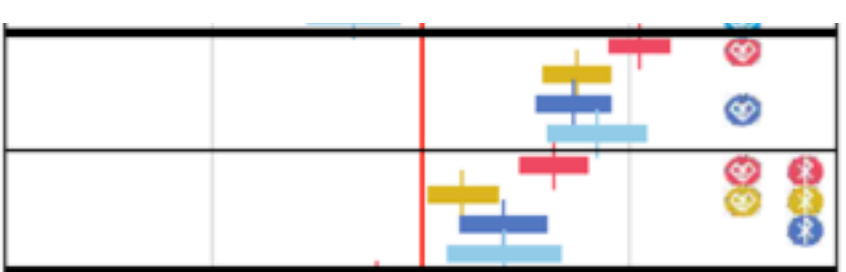
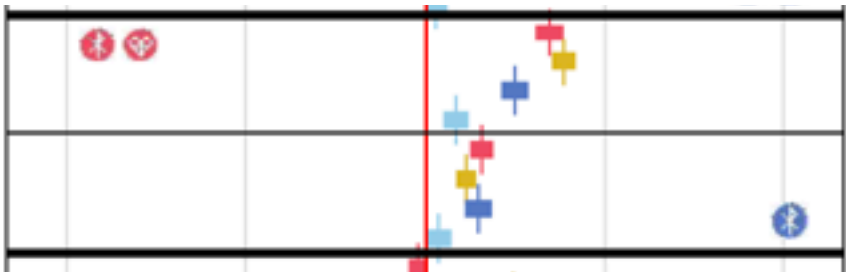
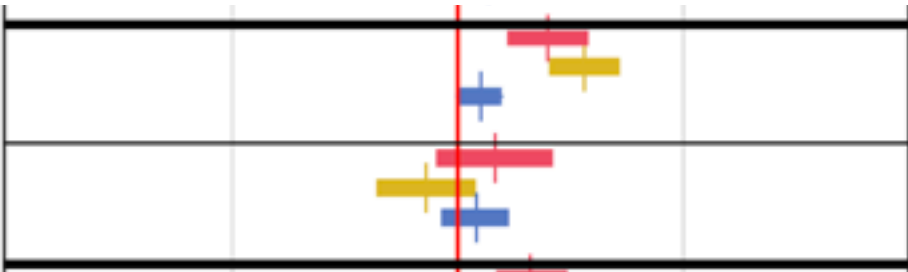
DAYS

ARGUMENTATION
STRATEGY



GROUP
RHETORIC

Excl.outgroup
Incl.in/both groups



EMOTIONS

Any group rhetoric (both in- and exclusive) increases hate speech.

THREE MAIN FINDINGS: OPINIONS - YES, GROUP RHETORIC - NO, EMOTIONS - NO.

REPLY PAIRS

DISCUSSIONS

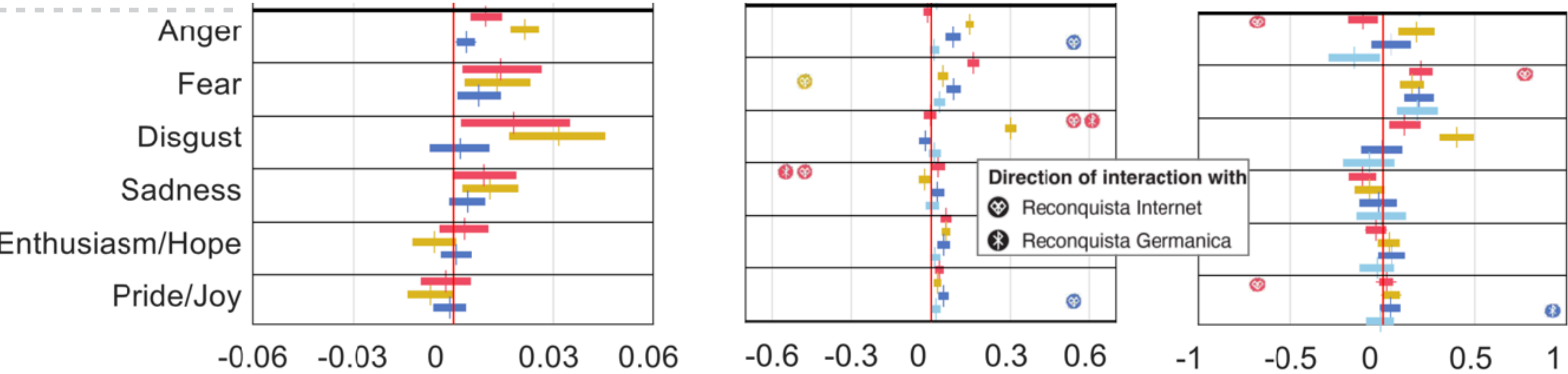
DAYS

ARGUMENTATION
STRATEGY

GROUP
RHETORIC

EMOTIONS

Almost all emotional speech makes
conversations more hateful.



THIS IS A GREAT STUDY, BECAUSE

- ▶ We provide the first large-scale observational study of counter speech in the wild.
- ▶ We use sophisticated mixed methods to measure hate and counter speech.
- ▶ We use matching and ARDL to provide quasi causal effects.

BUT IT ALSO HAS LIMITATIONS.

- ▶ The sample is specific to Germany and Twitter.
- ▶ Our study is observational and can thus be subject to unknown confounders.
- ▶ The observed effects are not always consistent across different levels and outcomes.

14:30 - 15:00

QUESTIONS. DISCUSSION.